

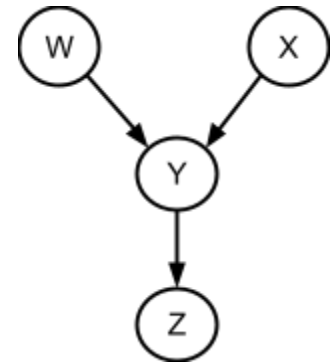
COMP/ELEC 440/557, Final Exam, Part 2

Consider the Bayesian network shown here with four scalar random variables: W , X , Y , and Z .

The prior probability distributions for W and X are known. Both are drawn from Gaussian distributions as described below, where μ denotes the mean of the distribution and σ its standard deviation.

$$W \sim \text{Gaussian}(\mu = 5, \sigma = 1)$$

$$X \sim \text{Gaussian}(\mu = 0, \sigma = 5)$$



The stochastic function for generating Z given Y is also known and available in the starter code as the function `def generate_z_given_y(y)` in `utils.py`.

However, the conditional probability distribution of Y is unknown. Instead, you have access to data of (W, X, Y) in the files named `training_data.csv` and `validation_data.csv`.

In Part 2, you will use AI methods to predict and infer value of variables within this Bayesian network, based on the values of other variables.

Code: The starter code is available in `P2_starter_code.zip` on Canvas.

Files to Edit and Submit: You will fill in portions of `pgm.py` during the assignment. Please *do not* change the other files in this distribution. You will submit `pgm.py` to **Final Exam: Part 2** on Gradescope.

Evaluation: Your code will be autograded for technical correctness and efficiency.

- Please do not change the names of any provided functions or classes within the code.
- Please ensure that your code is compatible with Python 3.10.
- You may only import the numpy package, packages from the [Python Standard Library](https://docs.python.org/3/library/index.html) (<https://docs.python.org/3/library/index.html>), and packages provided in the starter code. Importing any additional packages (such as pytorch, scikit-learn) is not allowed.
- To receive credit, please ensure that your code terminates within 30 minutes when evaluated using the Gradescope autograder. Efficient solutions take less than 5 minutes.
- Use of AI tools (e.g., ChatGPT, Gemini, Claude, Copilot) is not permitted.

Collaboration Policy: Collaborating on the final exam is not permitted.

Citation Policy: Please refer to the course syllabus.

Late Policy: Your answers to the final exam must be submitted on time. Auto-extensions cannot be used for the final exam. Please note that no credit will be given for late submissions.

Q1 (30 points). Predict Y given W and X

In this question, use the provided data in **training_data.csv** to develop a deterministic function that predicts Y based on W and X. You are required to implement the **train()** and **predict_y()** methods of the **PGM** class in the file **pgm.py**. If you choose to implement a neural network, you may build on the code provided in **nn.py** and **optim.py**

Your score will be determined by the accuracy of the **predict_y()** function on a held-out test set, which is hidden from you.

Hint: You can evaluate the accuracy of your function on a validation dataset (**validation_data.csv**) as follows:

```
python run.py --q1
```

Q2 (20 points). Infer W given Z

In this question, implement a function that estimates the mean and standard deviation of W given the value of Z. You should complete the **infer_w_given_z()** method of the **PGM** class. Your score will be based on the accuracy of this function on a test set, which is hidden from you.

Hint: We have provided two test cases for this question. You can verify if you are on the right track by using these test cases, which can be run as follows:

```
python run.py --q2
```

Q3 (30 points). Infer W given Z and X

In this question, implement a function that estimates the mean and standard deviation of W given the value of Z and X. You should complete the **infer_w_given_x_z()** method of the **PGM** class. Your score will be based on the accuracy of this function on a test set, which is hidden from you.

Hint: For this question too, two test cases are available. You can check if you are on the right track by running these test cases as follows:

```
python run.py --q3
```